Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	5
Topic	Three dimensional shapes II
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume
Theme	—
Curriculum objectives	• identify properties of cylinders and spheres • find the volume of a cube, a cuboid and a prism
Previous knowledge	• properties of cubes and cuboid • properties of pyramids and cones
Lesson sequence	• identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces. • find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes
Instructional materials	• shapes of cylinders and spheres.
Evaluation	• Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high. • Calculate the volume of cuboid with dimensions 5m by 2m by 3m. • What is the volume of a cube of 4cm × 4cm × 4cm. • A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank.

Lesson objectives

1.	identify properties of cylinders and spheres
2.	find the volume of a cube, a cuboid and a prism

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Three dimensional figures: properties of cylinders and spheres, and volume

Class: JSS1 **Subject:** Mathematics **Week:** 5

Section 1: Cylinders and Spheres: Getting to Know Their Shapes

Two shapes you see every day are the cylinder and the sphere. A tin of baked beans or a toilet roll is a cylinder; a football or an orange is a sphere. They are three-dimensional shapes, so they take up space, and each has its own special set of properties. Let us look at each one carefully.

The cylinder

Picture a tin of baked beans. A cylinder has two flat circular faces — one at the top and one at the bottom. These two circles are equal in size and lie parallel to each other. Between them runs a curved surface that joins the two circles, like the curved side of the tin. So a cylinder has:

two circular faces (the top and the bottom), equal and parallel;
one curved surface (the side);
no vertices (no sharp corner points);
two circular edges (the rims where each circular face meets the curved side).

If you count all the surfaces, a cylinder has 2 flat circular faces and 1 curved surface — 3 surfaces in all.

The sphere

Now picture a football. A sphere is perfectly round — every point on its surface is the same distance from its centre. A sphere has no flat faces, no edges and no corners. It is just one single curved surface that curves all the way around, with no starting or ending point.

Worked example

Is this statement true or false? “A cylinder has two circular faces and one curved surface.”

Answer: True. The two circular faces are the top and bottom of the cylinder, and the curved surface is the side that joins them. The statement describes the cylinder correctly.

Worked example

A ball has no flat faces, no edges and no vertices. Name the three-dimensional shape it is, and say how many surfaces it has.

Answer: The ball is a sphere. It has just one surface — a single curved surface that wraps all the way around it.

Practice

1. A solid cylinder has two circular faces (the top and the bottom) and one curved surface. How many faces does a cylinder have altogether?
2. How many circular faces does a solid cylinder have?
3. A sphere has no flat faces, no edges and no vertices. What is the name of its only surface?

Section 2: Finding the Volume of Cubes, Cuboids and Prisms

Volume tells you how much space a solid shape takes up. You measure it in cubic units, such as cubic centimetres (cm³) or cubic metres (m³). To find the volume of a cube or a cuboid, you multiply its length, its width and its height together.

Think of a cuboid as a stack of cubes. A cuboid that is 4 cm long, 2 cm wide and 3 cm high can be pictured as 3 layers, each layer holding $4 \times 2 = 8$ cubes. That gives $3 \times 8 = 24$ cubes, so its volume is 24 cm³. What you have done is simply multiply the three dimensions:

Volume of a cuboid = length × width × height

A cube is just a cuboid whose length, width and height are all the same size. So if a cube has edges of length s, its volume is:

Volume of a cube = s × s × s = s^3

A prism is a solid with two identical, parallel faces (its bases) joined by flat rectangular sides. The volume of any prism is found by multiplying the area of one base by the height of the prism:

Volume of a prism = area of base × height

A cuboid is itself a prism whose base is a rectangle, so its base area is length × width, and multiplying by the height gives exactly the same formula as before.

Worked example 1: the volume of a cuboid

Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.

Use Volume = length × width × height.

Volume = 8 × 4 × 3 = 32 × 3 = 96 cm^3.

So the cuboid takes up 96 cm^3 of space.

Worked example 2: the volume of a cube

Find the volume of a cube whose edges are each 4 cm long.

Use Volume = s^3 = s × s × s.

Volume = 4 × 4 × 4 = 16 × 4 = 64 cm^3.

So the cube takes up 64 cm^3 of space.

Worked example 3: the volume of a prism

A triangular prism has a base of area 12 cm^2 and a height of 5 cm. Find its volume.

Use Volume = area of base × height.

Volume = 12 × 5 = 60 cm^3.

So the prism takes up 60 cm^3 of space.

Practice

1. Find the volume of a cuboid that is 8 cm long, 4 cm wide and 3 cm high.
2. Find the volume of a cuboid with dimensions 5 m by 2 m by 3 m.
3. What is the volume of a cube of edges 4 cm × 4 cm × 4 cm?
4. A rectangular tank is 6 m long and 2 m wide and holds 36 m^3 of water when full. How deep is the water in the tank?

Check your work

Attempt every question before you read this page.

Cylinders and Spheres: Getting to Know Their Shapes

1. A cylinder has 2 faces (the two circular faces — the top and the bottom). The curved side is a curved surface, not counted as a flat face.

   The word "faces" means the flat circular surfaces. A cylinder has one at the top and one at the bottom, so it has 2 faces. The curved side is a curved surface, not a flat face, so it is not included in the count of faces.
2. A solid cylinder has 2 circular faces — one at the top and one at the bottom.
3. It is a curved surface — one single curved surface that wraps all the way around the sphere.

Finding the Volume of Cubes, Cuboids and Prisms

1. 96 cm^3

   Volume = length × width × height = 8 × 4 × 3 = 96 cm^3.
2. 30 m^3

   Volume = length × width × height = 5 × 2 × 3 = 30 m^3.

3. 64 cm^3

   Volume of a cube = $s^3 = 4 \times 4 \times 4 = 64\ cm^3$.

4. 3 m

   The tank is a cuboid, so Volume = length × width × depth. Here $36 = 6 \times 2 \times \text{depth} = 12 \times \text{depth}$. So depth $= 36 \div 12 = 3$ m.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-ab1f5c4b	**Date**	
Subtopic	Three dimensional figures: properties of cylinders and spheres, and volume	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____